



CINEMA TICKETING

Group : 5

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Class : 1CC4

CEP CCIT

FAKULTAS TEKNIK UNIVERSITAS INDONESIA

2025

PROJECT ON

CINEMA TICKETING

Developed by

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CINEMA TICKETING

Batch Code : ICC4

Start Date : 7/12/2025

End Date : 23/12/2025

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Names of Developer :

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Date of Submission: 22/12/2025

CERTIFICATE

This is to certify that this report titled “Cinema Ticketing” embodies the original work done by Alka Dewa, Arlow Devon Hargraves Takkas Sinaga, Alezuna M. Nadhif Pohan. Project in partial fulfillment of their course requirement at CEP-CCIT FTUI.

Coordinator:

M. Riza Iqbal Latief, S.T.

ACKNOWLEDGEMENT

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Finally, we appreciate all group members for their cooperation and teamwork in completing the Cinema Ticketing project.

BACKGROUND

Technology is developing rapidly and encourages businesses to use digital systems to improve efficiency and accuracy.

In the cinema industry, manual ticket booking can cause problems such as long queues, seat duplication, and data errors.

Therefore, a cinema ticketing system is needed to manage movie schedules, seat reservations, and ticket purchases in an organized and efficient way.

SYSTEM ANALYSIS

The cinema ticketing system is designed to help manage ticket sales and reservations digitally.

The system allows customers to view movie schedules, choose seats, and purchase tickets easily.

It uses a database named cinema_ticketing with several tables such as customer, movie, studio, schedule, seat number, ticket purchase, and reservation.

This system improves data accuracy, speeds up the ticketing process, and reduces manual errors.

DATABASE DESIGN

Database Name : cinema_ticketing

Number of Tables : 7

- customer
- movie
- studio
- schedule
- seat_number
- ticket_purchase
- reservation

Number of View : 4

- v_detail_reservasi;
- v_jadwal_film;
- v_riwayat_customer;
- v_kursi_terpesan

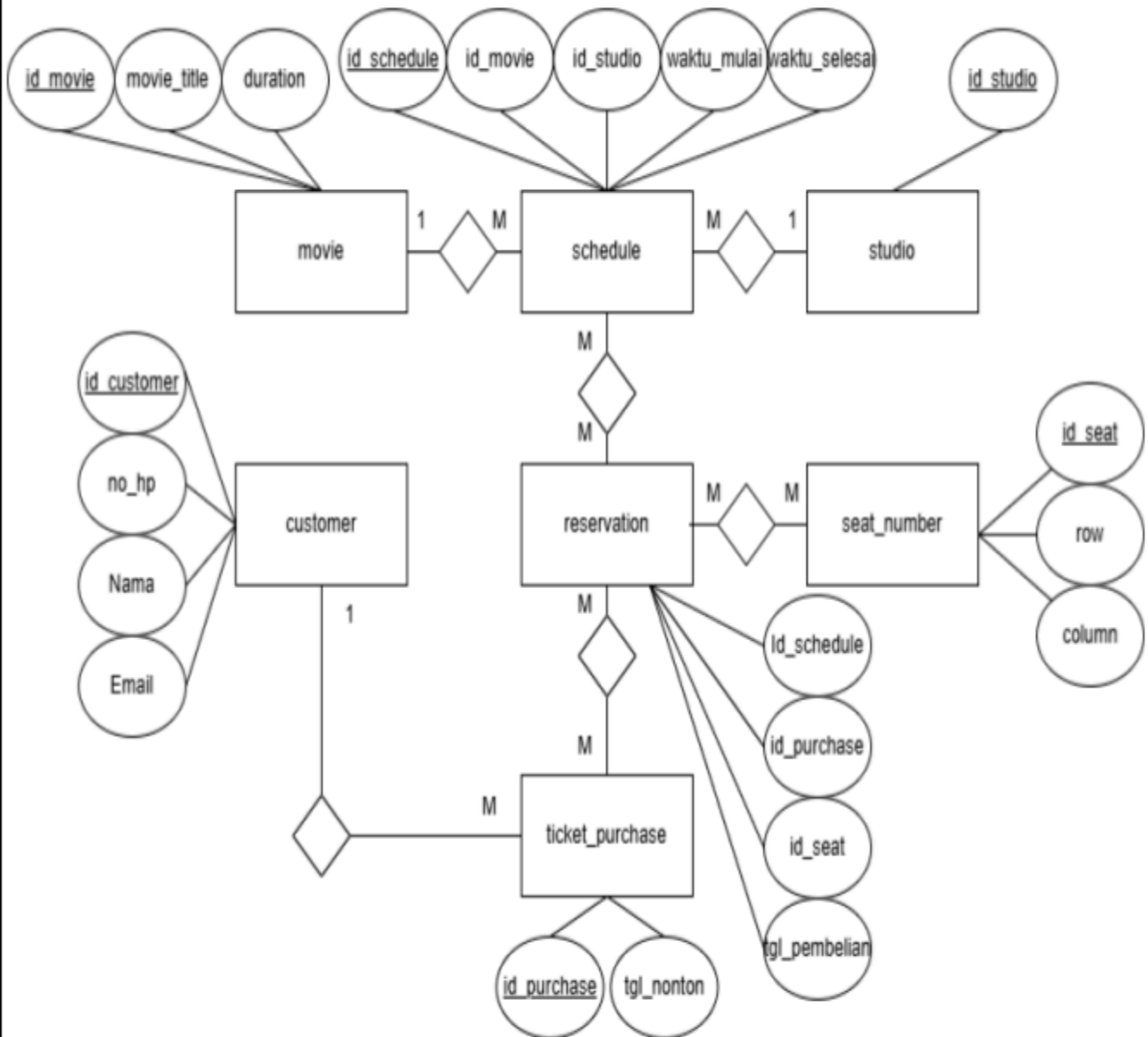
Number of Procedure: 5

- sp_detail_reservasi;
- sp_jadwal_film;
- sp_riwayat_customer;
- sp_tambah_reservasi;
- sp_kursi_terpesan;

Number of Trigger: 3

- trg_check_seat_studio;
- trg_check_double_booking;
- trg_set_tgl_pembelian;

ERD (Entity Relationship Diagram)



NORMALIZATION TABLES

NF 1

id_reservation	id_purchase	tgl_pembelian	tanggal_nonton	id_customer	nama	email	no_hp	id_movie	waktu_mulai	waktu_selesai	id_movie	movie_title	duration	id_studio	id_seat	row	column
R01	P01	2025-12-15	2025-12-20/2025	C01	Rian	rian@gmail.com	81293847645	SC01	13.00	15.00	SC01	zootopia	108 menit	S01	ST01	A	1
R02	P02	2025-12-15	2025-12-20/2025	C02	Alex	alex@gmail.com	82345689965	SC02	16.00	19.30	SC02	avengers	181 menit	S02	ST02	A	2
R03	P03	2025-12-16	2025-12-20/2025	C03	Tangguh	tangguh@gmail.com	85477357450	SC03	18.00	21.00	SC03	interstellar	169 menit	S03	ST03	B	1
R04	P04	2025-12-17	2025-12-17/2025	C04	Udin	udin@gmail.com	81356708912	SC04	18.00	20.00	SC04	zootopia	108 menit	S04	ST04	C	1
R05	P05	2025-12-12	2025-12-18/2025	C05	Asep	asep@gmail.com	85296884241	SC05	20.00	23.00	SC05	interstellar	169 menit	S05	ST05	C	2

NF 2

id_customer (PK)	nama	email	no_hp			
C01	Rian	rian@gmail.com	81293847645			
C02	Alex	alex@gmail.com	82345689965			
C03	Tangguh	tangguh@gmail.com	85477357450			
C04	Udin	udin@gmail.com	81356708912			
C05	Asep	asep@gmail.com	85296884241			

id_movie (PK)	movie_title	duration			
M01	zootopia	108 menit			
M02	avengers	181 menit			
M03	interstellar	169 menit			
M04	zootopia	108 menit			
M05	interstellar	169 menit			

id_schedule (PK)	id_movie (FK)	id_studio	waktu_mulai	waktu_selesai	
SCH01	M01	S01	13.00	15.00	
SCH02	M02	S02	16.00	19.30	
SCH03	M03	S03	18.00	21.00	
SCH04	M04	S04	18.00	20.00	
SCH05	M05	S05	20.00	23.00	

id_seat (PK)	row	column			
ST01	A	1			
ST02	A	2			
ST03	B	1			
ST04	C	1			
ST05	C	2			

id_reservation (PK)	id_purchase	tgl_pembelian	tanggal_nonton	id_customer (FK)	id_schedule (FK)	id_seat (FK)
R01	P01	12/15/2025	12/20/2025	C01	SCH01	ST01
R02	P02	12/15/2025	12/20/2025	C02	SCH02	ST02
R03	P03	12/16/2025	12/20/2025	C03	SCH03	ST03
R04	P04	12/10/2025	12/17/2025	C04	SCH04	ST04
R05	P05	12/12/2025	12/18/2025	C05	SCH05	ST05

NORMALIZATION TABLES

NF3

id_customer (PK)	nama	email	no_hp		
C01	Rian	rian@gmail.com	81293847845		
C02	Alex	alex@gmail.com	82345889965		
C03	Tangguh	tangguh@gmail.com	85477357450		
C04	Udin	udin@gmail.com	81356708912		
C05	Asep	asep@gmail.com	85298884241		
id_movie (PK)	movie_title	duration			
M01	zootopia	108 menit			
M02	avengers	181 menit			
M03	interstellar	169 menit			
id_schedule (PK)	id_movie (FK)	id_studio	waktu_mulai	waktu_selesai	
SCH01	M01	S01	13:00	15:00	
SCH02	M02	S02	16:00	19:30	
SCH03	M03	S03	18:00	21:00	
SCH04	M01	S04	18:00	20:00	
SCH05	M03	S05	20:00	23:00	
id_seat (PK)	row	column			
ST01	A	1			
ST02	A	2			
ST03	B	1			
ST04	C	1			
ST05	C	2			

Motivaskan yg pembelian dari detail kursi/bayar agar tidak terjadi redundansi jika satu transaksi memiliki banyak tiket.

id_reservation (PK)	id_purchase (FK)	id_schedule (FK)	id_seat (FK)	tanggal_nonton	
R01	P01	SCH01	ST01	12/20/2025	
R02	P02	SCH02	ST02	12/20/2025	
R03	P03	SCH03	ST03	12/20/2025	
R04	P04	SCH04	ST04	12/17/2025	
R05	P05	SCH05	ST05	12/18/2025	

Unnormalization

id_reservation	id_purchase	id_pembelian	tanggal_nonton	id_customer	nama	email	no_hp	id_schedule	waktu_mulai	waktu_selesai	id_movie	movie_title	duration	id_studio	id_seat	Row	Column
R01	P01	2025-12-15	2025-12-20	C01	Rian	rian@gmail.com	81293847845	SCH01	13:00	15:00	M01	zootopia	108 menit	S01	ST01	A	1
R02	P02	2025-12-15	2025-12-20	C02	Alex	alex@gmail.com	82345889965	SCH02	16:00	19:30	M02	avengers	181 menit	S02	ST02	A	2
R03	P03	2025-12-16	2025-12-20	C03	Tangguh	tangguh@gmail.com	85477357450	SCH03	18:00	21:00	M03	interstellar	169 menit	S03	ST03	B	1
R04	P04	2025-12-13	2025-12-17	C04	Udin	udin@gmail.com	81356708912	SCH04	18:00	20:00	M01	avengers	181 menit	S04	ST04	C	1
R05	P05	2025-12-12	2025-12-18	C05	Asep	asep@gmail.com	85298884241	SCH05	20:00	23:00	M03	zootopia	108 menit	S05	ST05	C	2

ERD DESIGN

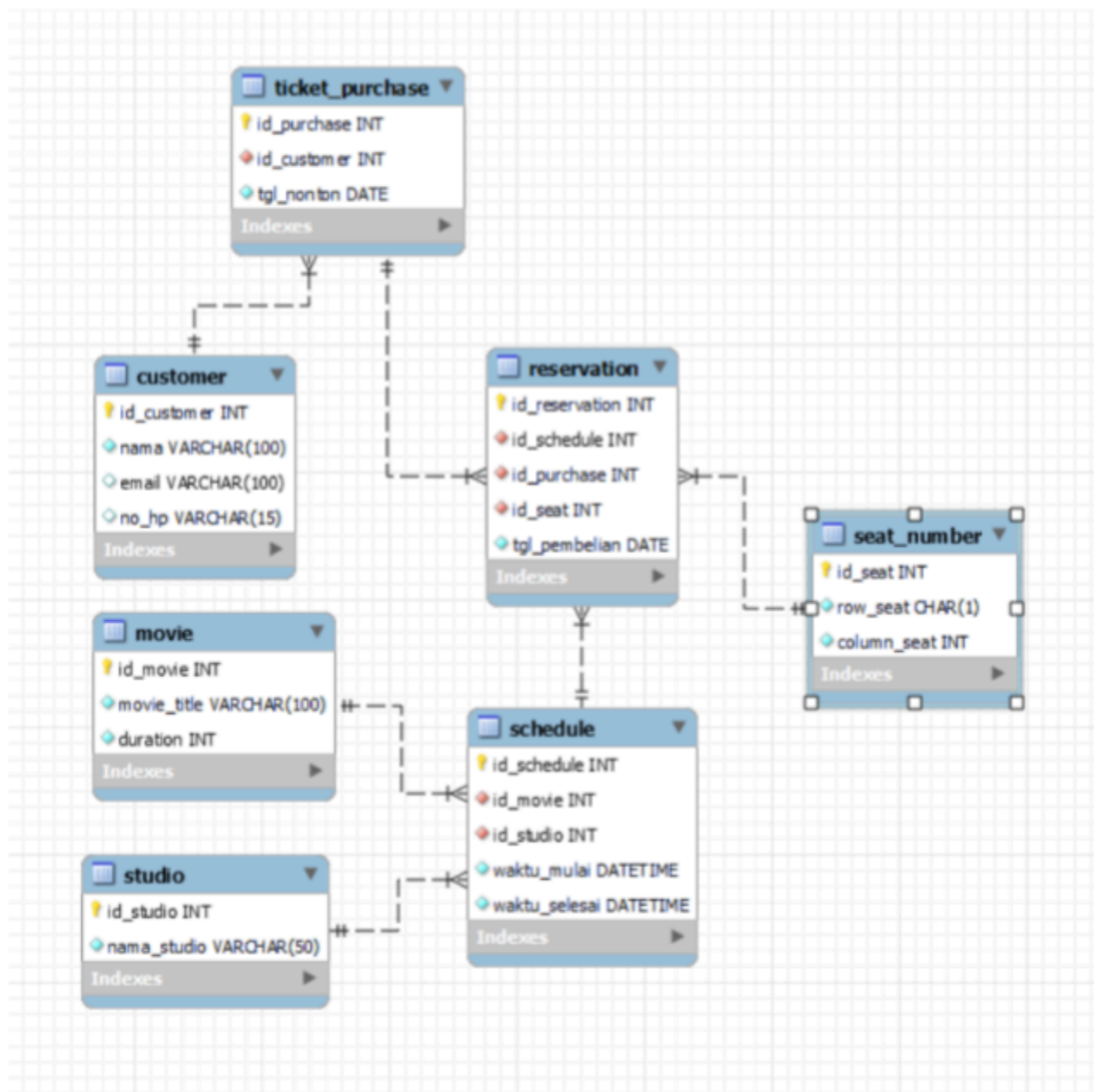


TABLE DESIGN

Customer Table

	Field	Type	Null	Key	Default	Extra
▶	id_customer	int	NO	PRI	NULL	auto_increment
	nama	varchar(100)	NO		NULL	
	email	varchar(100)	YES	UNI	NULL	
	no_hp	varchar(15)	YES		NULL	

Movie Table

	Field	Type	Null	Key	Default	Extra
▶	id_movie	int	NO	PRI	NULL	auto_increment
	movie_title	varchar(100)	NO		NULL	
	duration	int	NO		NULL	

Schedule Table

	Field	Type	Null	Key	Default	Extra
▶	id_schedule	int	NO	PRI	NULL	auto_increment
	id_movie	int	NO	MUL	NULL	
	id_studio	int	NO	MUL	NULL	
	waktu_mulai	datetime	NO		NULL	
	waktu_selesai	datetime	NO		NULL	

Seat Table

	Field	Type	Null	Key	Default	Extra
▶	id_seat	int	NO	PRI	NULL	auto_increment
	row_seat	char(1)	NO		NULL	
	column_seat	int	NO		NULL	

Studio Table

	Field	Type	Null	Key	Default	Extra
▶	id_studio	int	NO	PRI	NULL	auto_increment
	nama_studio	varchar(50)	NO		NULL	

TABLE DESIGN

Ticket Purchase Table

	Field	Type	Null	Key	Default	Extra
►	id_purchase	int	NO	PRI	NULL	auto_increment
	id_customer	int	NO	MUL	NULL	
	tgl_nonton	date	NO		NULL	

Reservation Table

	Field	Type	Null	Key	Default	Extra
►	id_reservation	int	NO	PRI	NULL	auto_increment
	id_schedule	int	NO	MUL	NULL	
	id_purchase	int	NO	MUL	NULL	
	id_seat	int	NO	MUL	NULL	
	tgl_pembelian	date	NO		NULL	

VALIDATION PERFORM

customer

Column	Method(s) for user validation	Description
id_customer (PK)	PRIMARY KEY, AUTO_INCREMENT, NOT NULL	This column contain Primary Key, cannot be empty
nama	NOT NULL	This column cannot be empty
email	NOT NULL	This column cannot be empty
no_hp	NOT NULL	This column cannot be empty

movie

Column	Method(s) for user validation	Description
id_movie (PK)	PRIMARY KEY, AUTO_INCREMENT, NOT NULL	This column contain Primary Key, cannot be empty
movie_title	NOT NULL	This column cannot be empty
duration	NOT NULL	This column cannot be empty

studio

Column	Method(s) for user validation	Description
id_studio (PK)	PRIMARY KEY, AUTO_INCREMENT, NOT NULL	This column contain Primary Key, cannot be empty
nama_studio	NOT NULL	This column cannot be empty

schedule

Column	Method(s) for user validation	Description
id_schedule (PK)	PRIMARY KEY, AUTO_INCREMENT, NOT NULL	This column contain Primary Key, cannot be empty
id_movie (FK)	FOREIGN KEY, NOT NULL	This column contain Foreign Key, references from movie table cannot be empty
id_studio (FK)	FOREIGN KEY, NOT NULL	This column contain Foreign Key, references from studio table cannot be empty
waktu_mulai	NOT NULL	This column cannot be empty
waktu_selesai	NOT NULL	This column cannot be empty

seat_number

Column	Method(s) for user validation	Description
id_seat (PK)	PRIMARY KEY, AUTO_INCREMENT, NOT NULL	This column contain Primary Key, cannot be empty
row_seat	NOT NULL	This column cannot be empty
column_seat	NOT NULL	This column cannot be empty

ticket_purchase

Column	Method(s) for user validation	Description
id_purchase (PK)	PRIMARY KEY, AUTO_INCREMENT, NOT NULL	This column contain Primary Key, cannot be empty
id_customer (FK)	FOREIGN KEY, NOT NULL	This column contain Foreign Key, references from customer table cannot be empty
tgl_nonton	NOT NULL	This column cannot be empty

reservation

Column	Method(s) for user validation	Description
id_reservation (PK)	PRIMARY KEY, AUTO_INCREMENT, NOT NULL	This column contain Primary Key, cannot be empty
id_schedule (FK)	FOREIGN KEY, NOT NULL	This column contain Foreign Key, references from schedule table cannot be empty
id_purchase (FK)	FOREIGN KEY, NOT NULL	This column contain Foreign Key, references from ticket_purchase table cannot be empty
id_seat (FK)	FOREIGN KEY, NOT NULL	This column contain Foreign Key, references from seat_number table cannot be empty
tgl_pembelian	NOT NULL	This column cannot be empty

CINEMA PROCEDURE

```
-- procedure Menampilkan detail tiket lengkap (customer, film,  
studio, jadwal, kursi)  
DELIMITER $$
```

```
CREATE PROCEDURE sp_detail_reservasi()  
BEGIN  
    SELECT  
        r.id_reservation,  
        c.nama AS nama_customer,  
        c.email,  
        m.movie_title,  
        st.nama_studio,  
        sch.waktu_mulai,  
        sch.waktu_selesai,  
        sn.row_seat,  
        sn.column_seat,  
        tp.tgl_nonton,  
        r.tgl_pembelian  
    FROM reservation r  
    JOIN ticket_purchase tp ON r.id_purchase = tp.id_purchase  
    JOIN customer c ON tp.id_customer = c.id_customer  
    JOIN schedule sch ON r.id_schedule = sch.id_schedule  
    JOIN movie m ON sch.id_movie = m.id_movie  
    JOIN studio st ON sch.id_studio = st.id_studio  
    JOIN seat_number sn ON r.id_seat = sn.id_seat;  
END$$
```

```
DELIMITER ;
```

```
-- CALL sp_detail_reservasi(); -
```

```
-- procedure Menampilkan jadwal film tertentu
```

```
DELIMITER $$
```

```
CREATE PROCEDURE sp_jadwal_film(  
    IN p_movie_title VARCHAR(100)  
)
```

CINEMA PROCEDURE

```
BEGIN
    SELECT
        m.movie_title,
        st.nama_studio,
        sch.waktu_mulai,
        sch.waktu_selesai
    FROM schedule sch
    JOIN movie m ON sch.id_movie = m.id_movie
    JOIN studio st ON sch.id_studio = st.id_studio
    WHERE m.movie_title = p_movie_title;
END$$

DELIMITER ;

-- CALL sp_jadwal_film('zootopia'); --

-- procedure Menampilkan pembelian berdasarkan id customer

DELIMITER $$

CREATE PROCEDURE sp_riwayat_customer(
    IN p_id_customer INT
)
BEGIN
    SELECT
        c.nama,
        c.email,
        tp.id_purchase,
        tp.tgl_nonton
    FROM ticket_purchase tp
    JOIN customer c ON tp.id_customer = c.id_customer
    WHERE c.id_customer = p_id_customer;
END$$

DELIMITER ;

-- CALL sp_riwayat_customer(1); --
```

CINEMA PROCEDURE

```
-- procedure Menambahkan data reservasi baru

DELIMITER $$

CREATE PROCEDURE sp_tambah_reservasi(
    IN p_id_schedule INT,
    IN p_id_purchase INT,
    IN p_id_seat INT
)
BEGIN
    INSERT INTO reservation (
        id_schedule,
        id_purchase,
        id_seat,
        tgl_pembelian
    )
    VALUES (
        p_id_schedule,
        p_id_purchase,
        p_id_seat,
        CURDATE()
    );
END$$

DELIMITER ;

-- CALL sp_tambah_reservasi(1, 1, 2); --

-- procedure Menampilkan kursi yang tidak tersedia

DELIMITER $$

CREATE PROCEDURE sp_kursi_terpesan(
    IN p_id_schedule INT
)
BEGIN
    SELECT
```

CINEMA PROCEDURE

```
        sn.row_seat,  
        sn.column_seat,  
        st.nama_studio  
FROM reservation r  
JOIN seat_number sn ON r.id_seat = sn.id_seat  
JOIN studio st ON sn.id_studio = st.id_studio  
WHERE r.id_schedule = p_id_schedule;  
END$$  
  
DELIMITER ;  
  
-- CALL kursi_terpesan(1); --
```

VIEW

-- View detail reservasi lengkap

```
CREATE VIEW v_detail_reservasi AS
SELECT
    r.id_reservation,
    c.id_customer,
    c.nama AS nama_customer,
    c.email,
    m.movie_title,
    st.nama_studio,
    sch.waktu_mulai,
    sch.waktu_selesai,
    sn.row_seat,
    sn.column_seat,
    tp.tgl_nonton,
    r.tgl_pembelian
FROM reservation r
JOIN ticket_purchase tp ON r.id_purchase = tp.id_purchase
JOIN customer c ON tp.id_customer = c.id_customer
JOIN schedule sch ON r.id_schedule = sch.id_schedule
JOIN movie m ON sch.id_movie = m.id_movie
JOIN studio st ON sch.id_studio = st.id_studio
JOIN seat_number sn ON r.id_seat = sn.id_seat;

-- SELECT * FROM v_detail_reservasi; --
```

-- view untuk melihat semua jadwal film

```
CREATE VIEW v_jadwal_film AS
SELECT
    sch.id_schedule,
    m.movie_title,
    st.nama_studio,
    sch.waktu_mulai,
    sch.waktu_selesai
FROM schedule sch
JOIN movie m ON sch.id_movie = m.id_movie
JOIN studio st ON sch.id_studio = st.id_studio;
```

VIEW

```
-- SELECT * FROM v_jadwal_film WHERE movie_title = 'zootopia'; --
```

```
-- view riwayat pembelian customer
```

```
CREATE VIEW v_riwayat_customer AS  
SELECT  
    c.id_customer,  
    c.nama,  
    c.email,  
    tp.id_purchase,  
    tp.tgl_nonton  
FROM ticket_purchase tp  
JOIN customer c ON tp.id_customer = c.id_customer;
```

```
-- SELECT * FROM v_riwayat_customer WHERE id_customer = 1; --
```

```
-- View untuk melihat kursi yang tidak tersedia
```

```
CREATE VIEW v_kursi_terpesan AS  
SELECT  
    r.id_schedule,  
    sn.id_seat,  
    sn.row_seat,  
    sn.column_seat,  
    st.nama_studio  
FROM reservation r  
JOIN seat_number sn ON r.id_seat = sn.id_seat  
JOIN studio st ON sn.id_studio = st.id_studio;
```

```
-- SELECT * FROM v_kursi_terpesan WHERE id_schedule = 1; --
```

TRIGGER

-- trigger Cek kursi harus sesuai studio jadwal

DELIMITER \$\$

```
CREATE TRIGGER trg_check_seat_studio
BEFORE INSERT ON reservation
FOR EACH ROW
BEGIN
```

```
    DECLARE v_studio_schedule INT;
    DECLARE v_studio_seat INT;
```

```
    -- Ambil studio dari jadwal
```

```
    SELECT id_studio
    INTO v_studio_schedule
    FROM schedule
    WHERE id_schedule = NEW.id_schedule;
```

```
    -- Ambil studio dari kursi
```

```
    SELECT id_studio
    INTO v_studio_seat
    FROM seat_number
    WHERE id_seat = NEW.id_seat;
```

```
    -- Validasi
```

```
    IF v_studio_schedule <> v_studio_seat THEN
        SIGNAL SQLSTATE '45000'
```

```
        SET MESSAGE_TEXT = 'Kursi tidak sesuai dengan studio
jadwal';
```

```
    END IF;
```

```
END$$
```

DELIMITER ;

-- trigger mencegah double booking

DELIMITER \$\$

```
CREATE TRIGGER trg_check_double_booking
```

TRIGGER

```
BEFORE INSERT ON reservation
FOR EACH ROW
BEGIN
    IF EXISTS (
        SELECT 1
        FROM reservation
        WHERE id_schedule = NEW.id_schedule
        AND id_seat = NEW.id_seat
    ) THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Kursi sudah dipesan pada jadwal ini';
    END IF;
END$$

DELIMITER ;

-- trigger otomatisasi input tanggal pembelian

DELIMITER $$

CREATE TRIGGER trg_set_tgl_pembelian
BEFORE INSERT ON reservation
FOR EACH ROW
BEGIN
    IF NEW.tgl_pembelian IS NULL THEN
        SET NEW.tgl_pembelian = CURDATE();
    END IF;
END$$

DELIMITER ;trg_check_seat_studio
```